ADVANCED CONTROL SYSTEMS

Manipulator Dynamics

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Outline

Dynamic model of robotic manipulators

Lagrange Formulation

Example: 1 DoF

Euler-Lagrange equations with constraints

Example: Spherical pendolum

PROJECT

Dynamic model of robotic manipulators

System theory studies the stability of systems like

$$\Sigma : \begin{cases} \dot{x}(t) = f(t, x(t), u(t)) \\ y(t) = h(t, x(t), u(t)) \end{cases} \qquad \Sigma : \begin{cases} \dot{x}(t) = Ax(t) + Bu(t)) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

and provides tools to design controllers able to

- stabilize the closed-loop system, and
- guarantee performance specifications

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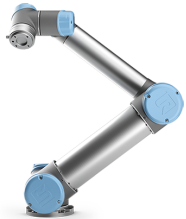
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"All models are wrong, but some are useful." –George E.P. Box

How can I compute a model for a robotic manipulator?

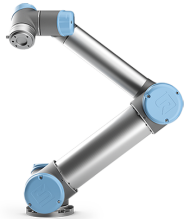








How can I compute a model for a robotic manipulator?

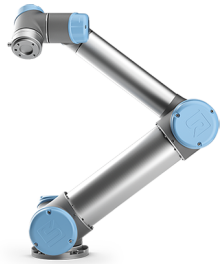








Why do I need a model?



The dynamical model mapping the generalized forces u into the generalized coordinates $q, \dot{q}$ will take the expression

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + F\dot{q} + g(q) = u$$

where

- $B(q)$ is the inertia tensor
- $C(q,\dot{q})\dot{q}$ is the Coriolis and centrifugal term
- $F\dot{q}$ is the friction
- $g(q)$ is the gravity term

The dynamical model is a *set of second-order nonlinear differential equations*

$$\mathcal{F}(q,\dot{q},\ddot{q}) = u$$

If $B(q)$ is nonsingular, the dynamical model

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + F\dot{q} + g(q) = u$$

can be written in the *state-space representation* $\dot{x}(t) = f(t, x, u)$ by defined the state vector x as

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \triangleq \begin{bmatrix} q \\ \dot{q} \end{bmatrix},$$

and re-arranging the implicit differential equations as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -B^{-1}(x_1)\left(C(x_1, x_2)x_2 + Fx_2 + g(x_1)\right) \end{bmatrix} + \begin{bmatrix} 0 \\ B^{-1}(x_1)u \end{bmatrix}$$

This formulation is useful to model the robot in Simulink

Another choice for the state vector is

$$\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \triangleq \begin{bmatrix} q \\ B(q)\dot{q} \end{bmatrix},$$

where the second half of $\bar{x}$ is the *generalized momentum*.

The state space model is

$$\begin{bmatrix} \dot{\bar{x}}_1 \\ \dot{\bar{x}}_2 \end{bmatrix} = \textbf{[Exercise]}$$

Exercise. Let

$$I\ddot{q} + F\dot{q} + mgd \sin q = \tau$$

be the model of 1-Dof link under gravity, where I is the inertia, m is the mass, d is the distance of the center of mass to the pivoting point, q is the angle with respect to the vertical axis (the same of the gravity q).

Write the state space model corresponding to the ODE.

Why the dynamic model is important

Derivation of the dynamic model of a manipulator plays an important role for

- simulation of motion,
- analysis of manipulator structures,
- design of control algorithms,
- design of prototype arms, joints, transmissions and actuators

The dynamic model provides the relationship between the joint actuator torques $u(t)$ and the robot motion, i.e. configuration $q(t)$.

We will start with the model in the joint space (i.e. q), later in operational space (i.e. x, *do not confuse this x with the state vector*)

1. Lagrange formulation (related to mechanical energy)
2. Newton-Euler formulation (recursive formulation)

Why the dynamic model is important

When a model is available it is possible to address the following problems:

- direct dynamics: $u \mapsto (q, \dot{q}, \ddot{q})$
- inverse dynamics: $(q, \dot{q}, \ddot{q}) \mapsto u$
- dynamic parameter identification

- motion control
- force control

- motion planning ($\rightarrow$ *Robotics, Vision and AI* course)
- visual servoing ($\rightarrow$ *Robotics, Vision and AI* course)
- Human-Robot Interaction / Cooperative Robots / Teleoperation ($\rightarrow$ *Physical Human-Robot Interaction* and *Medical Robotics* courses)

Dynamics taxonomy

Serial Robots (open kin chain)



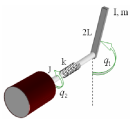
Parallel robots (closed kin chain)



Rigid-Joint–Rigid-Link Manipulators



Flexible-Joint–Rigid-Link Manipulators



Under-actuated robots

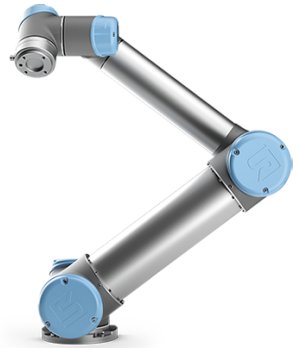


Over-actuated robots



inputs

$$\tau(t) = \begin{bmatrix} \tau_1(t) \\ \tau_2(t) \\ \vdots \\ \tau_n(t) \end{bmatrix}, \; t \in [0, T]$$



outputs

$$q(t) = \begin{bmatrix} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{bmatrix}, \; t \in [0, T]$$

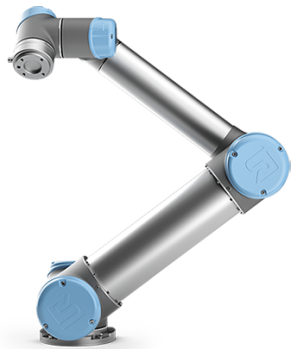
initial conditions:
$q(0), \dot{q}(0)$ at $t = 0$

Inverse dynamics

desired trajectory

$q_d(t)$, $\dot{q}_d(t)$ $\ddot{q}_d(t)$

$t \in [0, T]$



required commands

$\tau_d(t)$

$t \in [0, T]$

Here we assume that the desired trajectory $q_d(t)$ is known. ($\rightarrow$ *Robotics, Vision and AI* course)

Lagrange Formulation

Lagrangian function and Lagrange equation

Let $q \in \mathbb{R}^n$ be the generalized coordinates for a n-DOF manipulator (e.g. joint variables), $\mathcal{T}(q, \dot{q})$ be the kinetic energy, and $\mathcal{U}(q)$ be the potential energy

The Lagrangian of the mechanical system is

$$\mathcal{L}(q, \dot{q}) = \mathcal{T}(q, \dot{q}) - \mathcal{U}(q)$$

The Lagrange equations are

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T - \left(\frac{\partial \mathcal{L}}{\partial q}\right)^T = \tau$$

or, equivalently

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i, \qquad i = 1, \ldots, n$$

where τ_i are the generalized forces (external or dissipative).

The Lagrange equations establish the relations existing between the generalized forces applied to the manipulator and the joint positions, velocities and accelerations

The contributions to the generalized forces are given by the nonconservative forces, i.e., the joint actuator torques, the joint friction torques, as well as the joint torques induced by end-effector forces at the contact with the environment

Mechanical structure + parameters

$$\downarrow$$

$$\mathcal{T}(q, \dot{q}), \mathcal{U}(q)$$

$$\downarrow$$

Dynamics by solving Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T - \left(\frac{\partial \mathcal{L}}{\partial q}\right)^T = \tau$$

Why is the Lagrangian $\mathcal{L}(q, \dot{q}) = \mathcal{T}(q, \dot{q}) - \mathcal{U}(q)$ like that?

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Why are the Lagrange equations $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T - \left(\frac{\partial \mathcal{L}}{\partial q}\right)^T = \tau$ like that?

The answer is at the core of Geometric mechanics. We will just provide the intuition by studying the motion of the simplest mechanical system.

Newtonian mechanics

A *point mass* is an indealized zero-dimension object completely described by its position $q \in \mathbb{R}$ and its mass $m \in \mathbb{R}^+$.

The *Newton's law* for the motion of a point mass is

$$m\ddot{q} = F$$

where F is the total force acting on the point mass.

The *kinematic energy* is

$$\mathcal{T}(\dot{q}) = \frac{1}{2} m\dot{q}^2$$

Newtonian mechanics

A *Newtonian potential system* is

$$m\ddot{q} = -\frac{\partial \mathcal{U}}{\partial q}$$

where $\mathcal{U}(q)$ is the potential energy (real-valued function).

The *total energy* of a Newtonian potential system is

$$\mathcal{E} \triangleq \mathcal{T} + \mathcal{U}$$

Theorem (Conservation of energy)

In any Newtonian potential system, total energy is conserved.

Theorem

Every Newtonian potential system

$$m\ddot{q} = -\frac{\partial \mathcal{U}}{\partial q}$$

is equivalent to the Euler-Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = 0$$

for the Lagrangian

$$\mathcal{L}(q, \dot{q}) = \mathcal{T}(\dot{q}) - \mathcal{U}(q).$$

The domain of $\mathcal{L}$ is called the (velocity) phase space.

Proof.

Since $\mathcal{L}(q, \dot{q}) = \mathcal{T}(\dot{q}) - \mathcal{U}(q)$, we have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{q}} &= \frac{\partial \mathcal{T}}{\partial \dot{q}} = \frac{\partial}{\partial \dot{q}} \left(\frac{1}{2} m \dot{q}^2 \right) = m\dot{q} \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) &= \frac{d\, m\dot{q}}{dt} = m\ddot{q} \\ \frac{\partial \mathcal{L}}{\partial q} &= -\frac{\partial \mathcal{U}}{\partial q}. \end{aligned}$$

Finally

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \left(\frac{\partial \mathcal{L}}{\partial q} \right) = 0 \iff m\ddot{q} = -\frac{\partial \mathcal{U}}{\partial q}$$

□

The energy function for a Lagrangian $\mathcal{L}(q, \dot{q})$ is

$$\mathcal{E} = \frac{\partial \mathcal{L}}{\partial \dot{q}} \cdot \dot{q} - \mathcal{L}$$

Theorem

In any Lagrangian system, the energy function is conserved.

Proof.

$$\frac{d\mathcal{E}}{dt} = \ldots = 0$$

□



if there are external or friction forces, the Euler-Lagrange equations are equal to their sum $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = \tau$

Is the Lagrangian mechanics “just” a ri-formulation of the Newtonian mechanics?

Absolutely not!

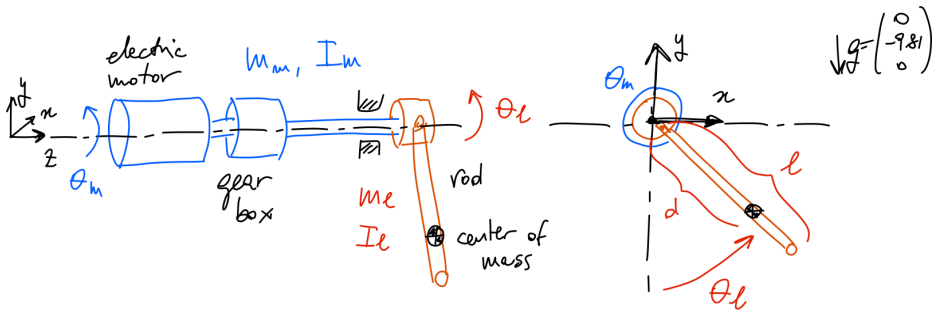
It is strongly related to many important results in Physics and Mathematics

- Variation principle
- Hamilton’s principle of stationary action
- Principle of Least Action
- D’Alambert principle
- Principle of Virtual works

and, in our context, it allows as to manage holonomic constraints (e.g. two links of a robotic manipulator connected by a joint)

Example: 1 DoF

Example 1



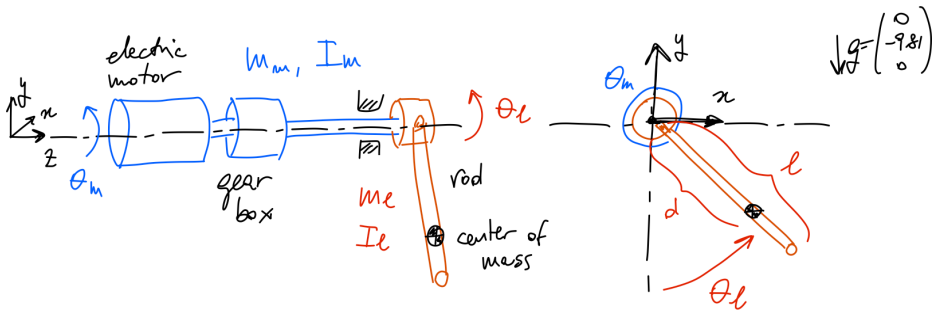
Relations between positions and torques

$$\theta_m = n\theta_\ell, \qquad \dot{\theta}_m = n\dot{\theta}_\ell, \qquad \tau_m = \frac{1}{n}\tau_\ell$$

Kinematic energy of the motor

$$\mathcal{T}_m = \frac{1}{2} I_m \dot{\theta}_m^2 = \frac{1}{2} I_m n^2 \dot{\theta}_\ell^2$$

Example 1



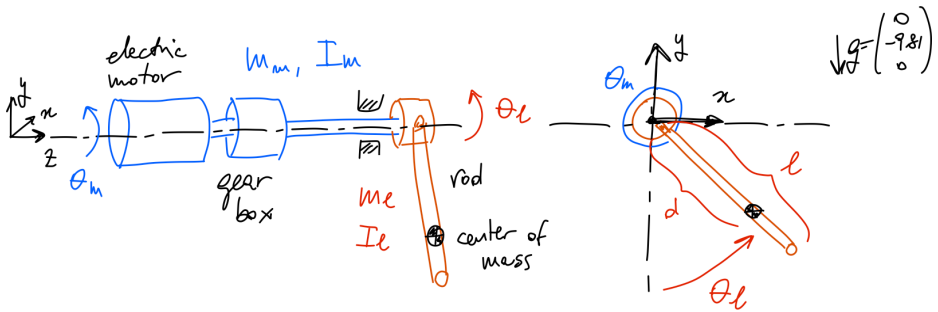
Kinematic energy of the link

$$\mathcal{T}_\ell = \frac{1}{2}(I_\ell + m_\ell d^2)\dot{\theta}_\ell^2 = \frac{1}{2}\bar{I}_\ell \dot{\theta}_\ell^2$$

Total Kinematic energy ($q = \theta_\ell$)

$$\mathcal{T} = \mathcal{T}_m + \mathcal{T}_\ell = \frac{1}{2} I_m n^2 \dot{\theta}_\ell^2 + \frac{1}{2}\bar{I}_\ell \dot{\theta}_\ell^2 = \frac{1}{2}\left(I_m n^2 + \bar{I}_\ell\right)\dot{\theta}_\ell^2 = \frac{1}{2} I\, \dot{\theta}_\ell^2 = \frac{1}{2} I\, \dot{q}^2$$

Example 1



Potential energy ($q = \theta_\ell$)

$$\mathcal{U} = \mathcal{U}_0 - mgd\cos\theta_l = \mathcal{U}_0 - mgd\cos q$$

Lagrangian function

$$\mathcal{L} = \mathcal{T} - \mathcal{U} = \frac{1}{2} I\dot{q}^2 - \mathcal{U}_0 + mgd\cos q$$

$$\mathcal{L}(q, \dot{q}) = \mathcal{T}(\dot{q}) - \mathcal{U}(q) = \frac{1}{2} I \dot{q}^2 - \mathcal{U}_0 + mgd\cos(q)$$

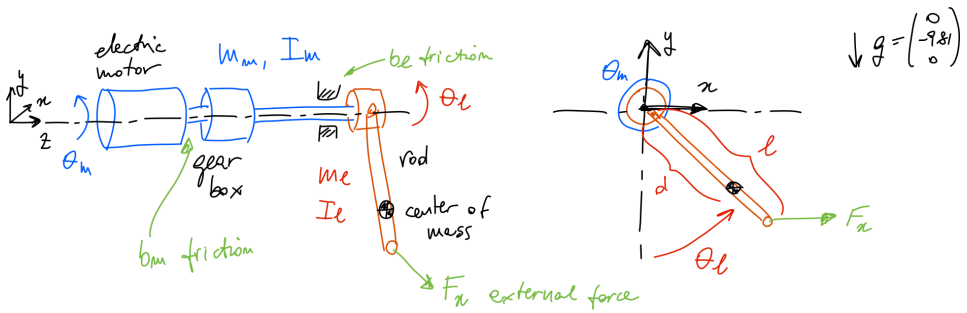
Terms within the Lagrange equation $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = \tau$, where we set $\tau = \tau_\ell$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{q}} &= I\dot{q} \\ \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) &= I\ddot{q} \\ \frac{\partial \mathcal{L}}{\partial q} &= -mgd\sin(q). \end{aligned}$$

Finally

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = \tau \iff I\ddot{q} + mgd\sin(q) = n\tau_m$$

Example 1

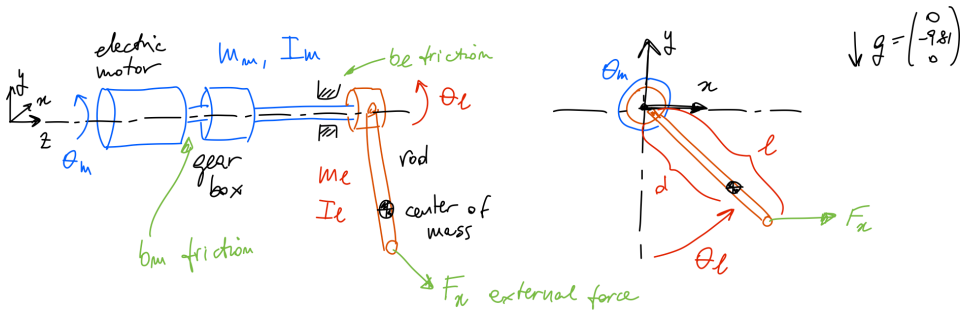


What happen with frictions ($b_m\dot{\theta}_m$, $b_\ell\dot{\theta}_\ell$)?

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = \tau - b_m n\dot{q} - b_\ell \dot{q}$$

$$I\ddot{q} + (b_m n + b_\ell)\dot{q} + mgd\sin(q) = n\tau_m$$

Example 1

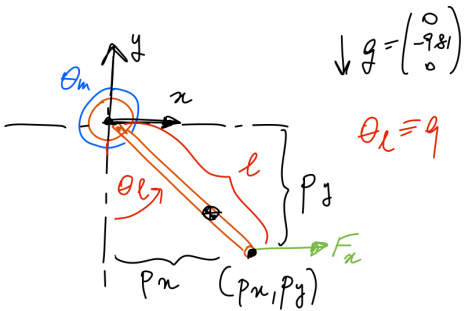


What happen with an external force along the x axix F_x?

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = \tau - b_m n \dot{q} - b_\ell \dot{q} + \ell \cos(q) F_x$$

$$I\ddot{q} + (b_m n + b_\ell)\dot{q} + mgd \sin(q) = n\tau_m + \ell \cos(q) F_x$$

Example 1



$$
\begin{aligned}
p_x &= \ell \sin(q) \\
p_y &= -\ell \cos(q) \\
\dot{p}_x &= \ell \cos(q)\, \dot{q} \\
\dot{p}_y &= \ell \sin(q)\, \dot{q} \\
\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \end{bmatrix} &= \underbrace{\begin{bmatrix} \ell \cos(q) \\ \ell \sin(q) \end{bmatrix}}_{J(q)} \dot{q}, \qquad J(q) \text{ Jacobian}
\end{aligned}
$$

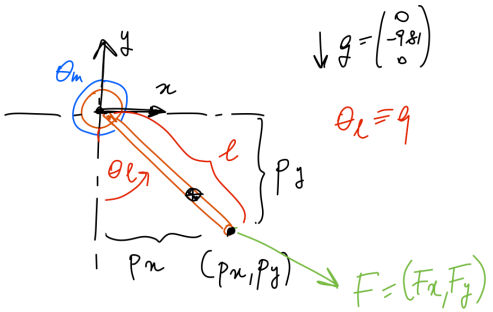
Relationship between external forces at the end-effector F_e and joint torques τ_e

$$\tau_e = J^T(q) F_e$$

i.e. equivalent joint torque τ_e due to the force F_e applied at the tip (p_x, p_y). In our case

$$F_e = \begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} F_x \\ 0 \end{bmatrix} \qquad \rightarrow \qquad \tau_e = \ell \cos(q) F_x$$

Example 1



$$
\begin{aligned}
p_x &= \ell \sin(q) \\
p_y &= -\ell \cos(q) \\
\dot{p}_x &= \ell \cos(q)\, \dot{q} \\
\dot{p}_y &= \ell \sin(q)\, \dot{q} \\
\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \end{bmatrix} &= \underbrace{\begin{bmatrix} \ell \cos(q) \\ -\ell \sin(q) \end{bmatrix}}_{J(q)} \dot{q},
\end{aligned}
\qquad J(q) \text{ Jacobian}
$$

With $F_e = \begin{bmatrix} F_x \\ F_y \end{bmatrix}$ we will have

$$
I\ddot{q} + (b_m n + b_\ell)\dot{q} + mgd \sin(q) = n\tau_m + \begin{bmatrix} \ell \cos(q) & -\ell \sin(q) \end{bmatrix} \begin{bmatrix} F_x \\ F_y \end{bmatrix}
$$

where the non-conservative torques are indicated in blue.

Example 1

Exercise. Re-write the dynamical model with $q = \theta_m$

Euler-Lagrange equations with constraints

Theorem

Every constrained Newtonian potential system

$$m_i \ddot{q}_i = -\frac{\partial \mathcal{U}}{\partial q_i} + C_i, \qquad i = 1, \ldots, N$$

with constraints $f_j(q) = c_j$, $j = 1, \ldots, k$ *and and constraint forces* C_i *satisfying*

$$(C_1, \ldots, C_N) = \sum_{j=1}^{k} \lambda_j \frac{\partial f_j(q)}{\partial q}$$

is equivalent to the Euler-Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = \sum_{j=1}^{k} \lambda_j \frac{\partial f_j(q)}{\partial q}$$

for the Lagrangian $\mathcal{L}(q, \dot{q}) = \mathcal{T}(\dot{q}) - \mathcal{U}(q)$.

Remark.
The Euler-Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \left(\frac{\partial \mathcal{L}}{\partial q}\right) = \sum_{j=1}^{k} \lambda_j \frac{\partial f_j(q)}{\partial q}$$

for the Lagrangian $\mathcal{L}(q, \dot{q}) = \mathcal{T}(\dot{q}) - \mathcal{U}(q)$ are equivalent to the the Euler-Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial \bar{\mathcal{L}}}{\partial \dot{q}}\right) - \left(\frac{\partial \bar{\mathcal{L}}}{\partial q}\right) = 0$$

for the Lagrangian $\bar{\mathcal{L}}(q, \dot{q}) = \mathcal{L}(q, \dot{q}) + \sum_{j=1}^{k} \lambda_j f_j(q)$.
Hint.

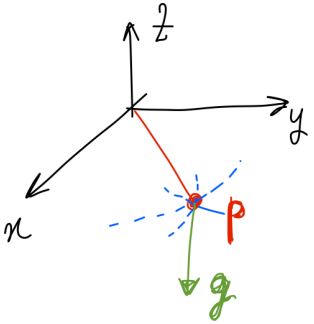
$$\frac{\partial \bar{\mathcal{L}}}{\partial q} = \frac{\partial \mathcal{L}}{\partial q} + \sum_{j=1}^{k} \lambda_j \frac{\partial f_j(q)}{\partial q}$$

The numbers λ_j are called *Lagrange multipliers*. To find explicit equations of motion in the variables q_i and $\dot{q}_i$ only, the Lagrange multipliers must be eliminated, which can be difficult.

The problem is much easier if the constraints have simple relationships with the coordinates $\rightarrow$ changes of coordinates

Example: Spherical pendolum

Example 2: Spherical pendolum



Spherical pendolum: a point mass of mass m, suspended from the origin by a massless rigid rod of length r

Constant gravitational force: $\boldsymbol{g} = -g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Constraint:

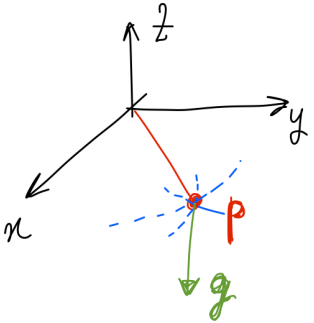
$$f(x, y, z) = r^2, \qquad f(x, y, z) \triangleq x^2 + y^2 + z^2$$

Configuration space:

$$Q = \{\boldsymbol{p} = (x, y, z) \in \mathbb{R}^3 | \, \|\boldsymbol{p}\| = r\}$$

i.e. the sphere of radius r

Example 2: Spherical pendolum



Unconstrained Lagrangian:

$$\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Constrained Lagrangian:

$$\begin{aligned} \bar{\mathcal{L}} &= \mathcal{L} - \lambda f \\ &\overset{(*)}{=} \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz - \lambda(x^2 + y^2 + z^2) \end{aligned}$$

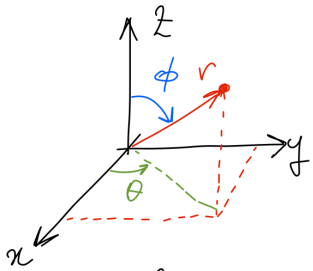
The motion is determined by the *Euler-Lagrange equations for* $\bar{L}$

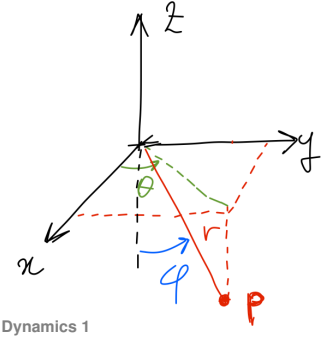
$$\begin{aligned} m\ddot{x} &= -2\lambda x \\ m\ddot{y} &= -2\lambda y \\ m\ddot{z} &= -2\lambda z - mg \end{aligned}$$

with the *constraint equation* $x^2 + y^2 + z^2 = r^2$

$(*)$: *The change of the sign in front of* λ *is just to make the following equations better looking.*

Example 2: Spherical pendolum





The EL equations are easier to solve in spherical coordinates

Cartesian coordinates $(x, y, z) \leftrightarrow (r, \phi, \theta)$ Spherical coordinates

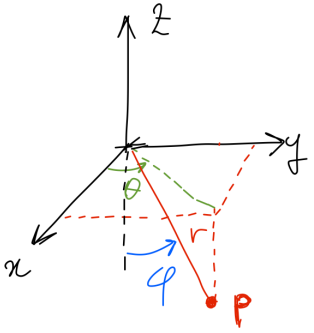
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r \sin\phi \cos\theta \\ r \sin\phi \sin\theta \\ r \cos\phi \end{bmatrix}$$

Cartesian coordinates $(x, y, z) \leftrightarrow (r, \varphi, \theta)$ Spherical coordinates

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r \sin\varphi \cos\theta \\ r \sin\varphi \sin\theta \\ -r \cos\varphi \end{bmatrix}$$

Constraint:

$$f(r, \varphi, \theta) = r^2, \qquad (r \text{ is constant here!})$$



Constrained Lagrangian:

$$\bar{\mathcal{L}} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2 + r^2\sin^2\varphi\,\dot{\theta}^2) + mgr\cos\varphi + \lambda r^2$$

Euler-Lagrange equations for $\bar{\mathcal{L}}$

$$\frac{d}{dt}(m\dot{r}) = mr\dot{\varphi}^2 + mr\sin^2(\varphi)\,\dot{\theta}^2 + mg\cos\varphi + 2\lambda r \quad (1)$$

$$\frac{d}{dt}(mr^2\dot{\varphi}) = mr^2\sin(\varphi)\cos(\varphi)\dot{\theta}^2 - mgr\sin\varphi \quad (2)$$

$$\frac{d}{dt}(mr^2\sin^2\varphi\,\dot{\theta}) = 0 \quad (3)$$

Since r is constant, in the first equation we have $\frac{d}{dt}(m\dot{r}) = 0$ and so such equation is irrelevant for determining the motion $(\theta(t), \varphi(t))$. Such dynamics is given by the second and third equations from where the Lagrange multiplier λ disappeared.

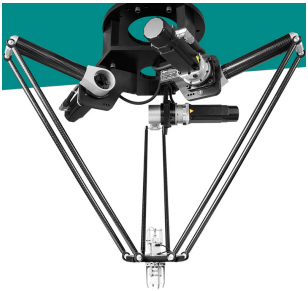
Euler-Lagrange equations with constraints

Exercise. Solve the EL equations (1)–(3) with Matlab

Exercise. Compute λ as a function of $\theta(t)$ and $\varphi(t)$ solutions of (2) and (3)

Exercise. Obtain the dynamic model of the Example 1 using the Lagrange multiplier

Remark. The Euler-Lagrange equations with constraints are needed to derive the dynamic model of parallel robots, like the Delta robot.



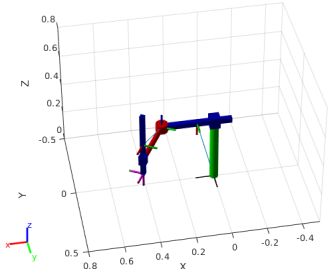

Project

A different 3DoF robot for each of you (example `PRP.urdf`)

This activity is a *personal work*: it is your exam !!! NO FOCUS GROUP

I give you the robot “shape”, you should choose the numerical parameters for the simulations

Example `PRP.urdf`





The reference frames in the Matlab plot of the URDF ARE NOT the frames related to the DH convention

PROJECT – Assignment # 1

To do

- DH table
- direct kinematics
- inverse kinematics
- Jacobians (geometric and analytical)

By hand, and cross-checking with Robotics toolbox